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*
*          STAAD.Pro
*          Version 2007 Build 04
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date= JUN 1, 2012
*          Time= 9:35:36
*
*          USER ID: Parsons Binckerhoff
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1. STAAD SPACE DXF IMPORT OF PR421-PR425.DXF
INPUT FILE: I20531_ST34 All MPS steel.STD
2. START JOB INFORMATION
3. ENGINEER DATE 14-MAY-12
4. JOB NAME JRC
5. END JOB INFORMATION
6. INPUT WIDTH 79
7. UNIT METER KN
8. JOINT COORDINATES
9. 1 1.2 0 0; 2 0 4.5 0; 3 3 0 0; 4 3 4.5 0; 5 3.8 4.5 0; 6 10.795 0 6.6
10. 7 10.795 3.69 6.6; 8 15.895 0 6.6; 9 15.895 3.69 6.6; 10 16.695 3.69 6.6
11. 11 9.995 3.69 6.6; 12 10.795 2.8 6.6; 13 8.395 2.8 6.6; 14 11.445 0 -0.2
12. 20 11.445 2.56 -0.2; 21 15.895 2.56 -0.2; 22 24 0 0; 23 24 4.5 0
13. 25 24.8 4.5 0; 27 23.2 4.5 0; 28 31 0 0; 29 31 5.377 0; 30 31.8 5.377 0
14. 31 30.2 5.377 0; 32 6.49 0 5.7; 33 6.49 6.5 5.7; 34 6.49 6.5 6.4
15. 35 6.49 6.5 5; 36 37 0 0; 37 37 6.5 0; 40 40 0 0; 41 40 6.5 0; 42 40.8 6.5 0
16. 43 44.7 0 0; 44 44.7 4.723 0; 45 45.5 4.723 0; 46 43.9 4.723 0; 47 0.9 4.5 0
17. 48 1.5 4.5 0; 49 2.1 4.5 0; 50 23.7 4.5 0; 51 24.3 4.5 0; 52 12.495 3.69 6.6
18. 53 24 0 8; 54 24 4.5 8; 55 24.8 4.5 8; 56 23.2 4.5 8; 57 23.7 4.5 8
19. 58 24.3 4.5 8; 59 37 0 3.3; 60 37 6.5 3.3; 62 40 0 3.3; 63 40 6.5 3.3
20. 64 40.8 6.5 3.3; 65 37 3.25 0; 66 40 3.25 0; 67 37 3.25 3.3; 68 40 3.25 3.3
21. 69 14.035 2.56 -0.2; 70 37.7 6.5 0; 71 45 4.723 0; 72 44.4 4.723 0
22. 73 9.195 2.8 6.6; 74 10.795 1.2 6.6; 75 1.2 0 1.5; 76 3 0 1.5; 77 1.2 2 0
23. 78 3 2 0; 79 24 3.5 0; 80 24 3.5 8; 81 24 4.5 1; 82 24 4.5 7; 84 6.49 2.8 5.7
24. 85 10.795 0 5.1; 86 10.795 3.69 5.1; 87 15.895 0 5.1; 88 15.895 3.69 5.1
25. 89 16.695 3.69 5.1; 90 9.995 3.69 5.1; 91 10.795 2.8 5.1; 92 8.395 2.8 5.1
26. 93 12.495 3.69 5.1; 94 9.195 2.8 5.1; 95 10.795 1.2 5.1; 96 15.895 0 -1.7
27. 98 15.895 2.56 -1.7; 99 15.895 0 1.3; 100 15.895 3.69 1.3; 101 15.895 2.56 1.3
28. 192 15.895 3.69 -0.2; 193 11.445 3.69 -0.2; 194 16.695 3.69 -0.2
29. 195 10.645 3.69 -0.2; 196 15.895 3.69 -1.7; 197 11.445 3.69 5.1
30. 198 12.495 3.69 -0.2; 199 1.2 4.5 0; 210 8.395 2.8 5.7
31. MEMBER INCIDENCES
32. 1 2 47; 2 4 5; 3 1 77; 4 3 78; 5 7 52; 6 9 10; 7 6 74; 8 8 9; 9 11 7; 10 12 7
33. 11 13 73; 14 14 20; 19 20 69; 20 23 51; 22 22 79; 24 27 50; 25 29 30; 26 28 29
34. 27 31 29; 28 32 84; 29 35 33; 30 33 34; 31 36 65; 33 37 70; 34 40 66; 36 41 42
35. 37 43 44; 38 46 72; 39 44 71; 40 47 199; 41 48 49; 42 49 4; 43 50 23; 44 51 25
36. 46 52 9; 48 54 58; 49 53 80; 50 56 57; 51 57 54; 52 58 55; 53 23 81; 54 59 67
37. 56 60 63; 57 62 68; 58 63 64; 59 37 60; 60 41 63; 61 65 37; 62 66 41; 63 67 60
38. 64 68 63; 65 65 67; 66 66 68; 67 36 67; 68 59 65; 69 65 60; 70 67 37; 71 66 63
39. 72 68 41; 73 40 68; 74 62 66; 75 69 21; 76 70 41; 77 71 45; 78 72 44; 79 73 12
40. 80 74 12; 81 74 73; 83 77 199; 84 78 4; 85 75 77; 86 76 78; 87 79 23; 88 80 54
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41. 89 81 82; 90 82 54; 91 79 81; 92 80 82; 93 84 33; 95 86 197; 96 88 89
42. 97 85 95; 98 87 88; 99 90 86; 100 91 86; 101 92 94; 102 93 88; 103 94 91
43. 104 95 91; 105 95 94; 106 86 7; 107 88 9; 108 92 210; 110 92 12; 112 8 88
44. 113 87 9; 114 85 7; 115 6 86; 116 96 98; 118 99 101; 119 101 100; 122 98 21
45. 123 21 101; 202 100 88; 221 192 100; 222 193 198; 223 192 194; 224 195 193
46. 227 20 193; 228 21 192; 229 193 197; 231 98 196; 232 196 192; 233 197 93
47. 234 198 192; 235 199 48; 249 210 13; 250 84 210; 256 91 13
48. DEFINE MATERIAL START
49. ISOTROPIC STEEL
50. E 2.05E+008
51. POISSON 0.3
52. DENSITY 76.8195
53. ALPHA 1.2E-005
54. DAMP 0.03
55. ISOTROPIC CONCRETE
56. E 2.17185E+007
57. POISSON 0.17
58. DENSITY 23.5616
59. ALPHA 1E-005
60. DAMP 0.05
61. END DEFINE MATERIAL
62. MEMBER PROPERTY JAPANESE
63. 85 86 91 92 108 202 221 228 229 232 249 250 TABLE ST H200X200X8
64. 11 19 59 60 65 66 75 79 81 101 103 105 TO 107 TABLE ST H200X200X8
65. 67 TO 74 112 TO 115 TABLE ST C150X75X6.5
66. 1 TO 10 14 28 TO 31 34 40 TO 42 46 54 57 61 TO 64 80 83 84 93 95 TO 100 102 -
67. 104 116 118 119 122 123 222 TO 224 227 231 233 TO 235 TABLE ST H300X300X10
68. 20 22 24 43 44 48 TO 53 87 TO 90 TABLE ST H250X250X9
69. 37 TO 39 77 78 TABLE ST H350X350X12
70. 110 256 TABLE ST L75X75X9
71. MEMBER PROPERTY JAPANESE
72. 33 36 56 58 76 TABLE ST H440X300X11
73. 25 TO 27 TABLE ST H394X398X11
74. CONSTANTS
75. BETA 90 MEMB 26 37 116 118 119 231
76. MATERIAL STEEL ALL
77. SUPPORTS
78. 14 22 28 32 43 53 96 99 FIXED
79. 1 3 6 8 36 59 62 75 76 85 87 PINNED
80. 40 FIXED BUT MY MZ KMX 0.001
81. MEMBER TENSION
82. 67 TO 74 110 112 TO 115 256
83. MEMBER RELEASE
84. 59 60 65 66 81 105 TO 108 202 221 229 232 250 START MY MZ
85. 59 60 65 66 75 81 105 TO 107 202 221 229 232 249 250 END MY MZ
86. 90 END MY MZ
87. 53 START MY MZ
88. 19 START MY MZ
89. LOAD 1 LOADTYPE NONE TITLE DL
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
MASTER/SLAVE OR IF UNCONNECTED JOINTS.
90. SELFWEIGHT Y -1 LIST 1 TO 11 14 19 20 22 24 TO 31 33 34 36 TO 44 46 -
91. 48 TO 54 56 TO 81 83 TO 93 95 TO 108 110 112 TO 116 118 119 122 123 202 221 -
92. 222 TO 224 227 TO 229 231 TO 235 249 250 256
93. JOINT LOAD
94. 47 TO 49 71 72 FY -28
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95. 4 FY -77
96. 7 9 41 52 70 198 FY -116.3
97. 192 FY -87.2
98. 69 FY -33.7
99. 50 51 57 58 FY -22.4
100. 29 FY -96.2
101. MEMBER LOAD
102. 29 30 UNI GY -25
103. 11 UNI GY -28
104. 79 UNI GY -28 0 0.8
105. LOAD 2 LOADTYPE NONE TITLE FL
106. JOINT LOAD
107. 47 TO 49 71 72 FZ 2.8
108. 4 FZ 7.7
109. 7 9 41 52 70 198 FZ 11.7
110. 192 FZ 8.8
111. 69 FZ 3.4
112. 50 51 57 58 FZ 2.3
113. 29 FZ 9.7
114. MEMBER LOAD
115. 29 30 UNI GX 2.5
116. 11 UNI GZ 2.8
117. 79 UNI GZ 2.8 0 0.8
118. LOAD 3 LOADTYPE NONE TITLE WLX
119. MEMBER LOAD
120. 3 4 7 8 10 14 22 26 31 34 37 49 53 54 57 65 66 80 83 TO 90 97 98 100 104 106 -
121. 107 116 118 119 122 123 202 221 227 231 232 UNI GX 0.33
122. 67 68 73 74 UNI GX 0.17
123. 28 TO 30 59 TO 64 93 UNI GX 0.38
124. 56 69 TO 71 UNI GX 0.19
125. JOINT LOAD
126. 2 FX 4.9
127. 7 FX 5.1
128. 193 FX 3.5
129. 23 54 FX 1.9
130. 29 FX 4.1
131. 33 FX 1
132. 37 FX 5.5
133. 44 FX 2.4
134. LOAD 4 LOADTYPE NONE TITLE WLZ
135. MEMBER LOAD
136. 1 TO 8 10 11 14 19 20 22 24 TO 27 31 34 37 TO 44 46 48 TO 52 54 57 75 -
137. 77 TO 80 83 84 87 88 116 118 119 222 TO 224 227 231 234 235 UNI GZ 0.33
138. 28 33 36 56 58 61 TO 64 76 93 UNI GZ 0.38
139. LOAD 5 LOADTYPE NONE TITLE NLX
140. LOAD 6 LOADTYPE NONE TITLE NLZ
141. LOAD 7 LOADTYPE NONE TITLE 1.0DL+1.0FL
142. REPEAT LOAD
143. 1 1.0 2 1.0
144. LOAD 8 LOADTYPE NONE TITLE 1.0DL+1.0FL+1.0WLX
145. REPEAT LOAD
146. 1 1.0 2 1.0 3 1.0
147. LOAD 9 LOADTYPE NONE TITLE 1.0DL+1.0FL+1.0WLZ
148. REPEAT LOAD
149. 1 1.0 2 1.0 4 1.0
150. LOAD 10 LOADTYPE NONE TITLE 1.0DL+1.0FL+1.0NLX

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151. REPEAT LOAD
152. 1 1.0 2 1.0 5 1.0
153. LOAD 11 LOADTYPE NONE TITLE 1.0DL+1.0FL+1.0NLZ
154. REPEAT LOAD
155. 1 1.0 2 1.0 6 1.0
156. LOAD 12 LOADTYPE NONE TITLE 1.0DL-1.0FL
157. REPEAT LOAD
158. 1 1.0 2 -1.0
159. LOAD 13 LOADTYPE NONE TITLE 1.0DL-1.0FL +1.0WLX
160. REPEAT LOAD
161. 1 1.0 2 -1.0 3 1.0
162. LOAD 14 LOADTYPE NONE TITLE 1.0DL-1.0FL-1.0WLX
163. REPEAT LOAD
164. 1 1.0 2 -1.0 3 -1.0
165. LOAD 15 LOADTYPE NONE TITLE 1.0DL-1.0FL+1.0WLZ
166. REPEAT LOAD
167. 1 1.0 2 -1.0 4 1.0
168. LOAD 16 LOADTYPE NONE TITLE 1.0DL-1.0FL-1.0WLZ
169. REPEAT LOAD
170. 1 1.0 2 -1.0 4 -1.0
171. LOAD 17 LOADTYPE NONE TITLE 1.0DL-1.0FL+1.0 NLX
172. REPEAT LOAD
173. 1 1.0 2 -1.0 5 1.0
174. LOAD 18 LOADTYPE NONE TITLE 1.0DL-1.0FL-1.0 NLX
175. REPEAT LOAD
176. 1 1.0 2 -1.0 5 -1.0
177. LOAD 19 LOADTYPE NONE TITLE 1.0DL-1.0FL+1.0 NLZ
178. REPEAT LOAD
179. 1 1.0 6 1.0 2 -1.0
180. LOAD 20 LOADTYPE NONE TITLE 1.0DL-1.0FL-1.0 NLZ
181. REPEAT LOAD
182. 2 -1.0 6 -1.0 1 1.0
183. LOAD 21 LOADTYPE NONE TITLE 1.0DL+1.0FL-1.0 WLX
184. REPEAT LOAD
185. 1 1.0 2 1.0 3 -1.0
186. LOAD 22 LOADTYPE NONE TITLE 1.0DL+1.0FL-1.0 WLZ
187. REPEAT LOAD
188. 4 -1.0 1 1.0 2 1.0
189. LOAD 23 LOADTYPE NONE TITLE 1.0DL+1.0FL-1.0 NLX
190. REPEAT LOAD
191. 1 1.0 2 1.0 5 -1.0
192. LOAD 24 LOADTYPE NONE TITLE 1.0DL+1.0FL-1.0 NLZ
193. REPEAT LOAD
194. 6 -1.0 1 1.0 2 1.0
195. *ULTIMATE LOAD COMBINATIONS
196. LOAD 25 LOADTYPE NONE TITLE 1.0DL+1.4FL
197. REPEAT LOAD
198. 1 1.0 2 1.4
199. LOAD 26 LOADTYPE NONE TITLE 1.4DL+1.4FL
200. REPEAT LOAD
201. 1 1.4 2 1.4
202. LOAD 27 LOADTYPE NONE TITLE 1.4DL+1.4FL+1.4WLX
203. REPEAT LOAD
204. 1 1.4 2 1.4 3 1.4
205. LOAD 28 LOADTYPE NONE TITLE 1.4DL+1.4FL+1.4WLZ
206. REPEAT LOAD

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207. 1 1.4 2 1.4 4 1.4
208. LOAD 29 LOADTYPE NONE TITLE 1.4DL+1.4FL+1.4NLX
209. REPEAT LOAD
210. 1 1.4 2 1.4 5 1.4
211. LOAD 30 LOADTYPE NONE TITLE 1.4DL+1.4FL+1.4NLZ
212. REPEAT LOAD
213. 1 1.4 2 1.4 6 1.4
214. LOAD 31 LOADTYPE NONE TITLE 1.0DL-1.4FL
215. REPEAT LOAD
216. 1 1.0 2 -1.4
217. LOAD 32 LOADTYPE NONE TITLE 1.4DL-1.4FL
218. REPEAT LOAD
219. 1 1.4 2 -1.4
220. LOAD 33 LOADTYPE NONE TITLE 1.4DL-1.4FL+1.4 WLX
221. REPEAT LOAD
222. 1 1.4 3 1.4 2 -1.4
223. LOAD 34 LOADTYPE NONE TITLE 1.4DL-1.4FL-1.4 WLX
224. REPEAT LOAD
225. 2 -1.4 3 -1.4 1 1.4
226. LOAD 35 LOADTYPE NONE TITLE 1.4DL-1.4FL+1.4 WLZ
227. REPEAT LOAD
228. 1 1.4 4 1.4 2 -1.4
229. LOAD 36 LOADTYPE NONE TITLE 1.4DL-1.4FL-1.4 WLZ
230. REPEAT LOAD
231. 2 -1.4 4 -1.4 1 1.4
232. LOAD 37 LOADTYPE NONE TITLE 1.4DL-1.4FL+1.4 NLX
233. REPEAT LOAD
234. 1 1.4 5 1.4 2 -1.4
235. LOAD 38 LOADTYPE NONE TITLE 1.4DL-1.4FL-1.4 NLX
236. REPEAT LOAD
237. 2 -1.4 5 -1.4 1 1.4
238. LOAD 39 LOADTYPE NONE TITLE 1.4DL-1.4FL+1.4 NLZ
239. REPEAT LOAD
240. 1 1.4 6 1.4 2 -1.4
241. LOAD 40 LOADTYPE NONE TITLE 1.4DL-1.4FL-1.4 NLZ
242. REPEAT LOAD
243. 2 -1.4 6 -1.4 1 1.4
244. LOAD 41 LOADTYPE NONE TITLE 1.4DL+1.4FL-1.4 WLX
245. REPEAT LOAD
246. 1 1.4 2 1.4 3 -1.4
247. LOAD 42 LOADTYPE NONE TITLE 1.4DL+1.4FL-1.4 WLZ
248. REPEAT LOAD
249. 4 -1.4 1 1.4 2 1.4
250. LOAD 43 LOADTYPE NONE TITLE 1.4DL+1.4FL-1.4 NLX
251. REPEAT LOAD
252. 1 1.4 2 1.4 5 -1.4
253. LOAD 44 LOADTYPE NONE TITLE 1.4DL+1.4FL-1.4 NLZ
254. REPEAT LOAD
255. 6 -1.4 1 1.4 2 1.4
256. PERFORM ANALYSIS

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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 98/ 119/ 20

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 85/ 8/ 51 DOF
TOTAL PRIMARY LOAD CASES = 44, TOTAL DEGREES OF FREEDOM = 504
SIZE OF STIFFNESS MATRIX = 26 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.9/ 0.0 MB

**** WARNING : AVAILABLE HARD DISK SPACE MAY NOT BE
ENOUGH TO COMPLETE EXECUTION. IF YOUR AVAILABLE HARD DISK
SPACE ON THE ANALYSIS DRIVE IS GREATER THAN 3GB THIS MESSAGE
MAY BE ERRONEOUS

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 1

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 2

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 3

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 4

**NOTE-Tension/Compression converged after 1 iterations, Case= 5

**NOTE-Tension/Compression converged after 1 iterations, Case= 6

**START ITERATION NO. 2